AutoSOC Cyber Analyst (ASOC-CA): Using AI to Automate SOC Tier 1 & 2 Activities

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***Abstract*—Hackers are now beginning to use artificial intelligence (AI) to expand the capabilities and sophistication of their attacks, creating challenges for network owners. As state-sponsored adversaries and advanced persistent threats (APTs) utilize AI to their advantage, network owners must evolve concurrently to address the emergent threat landscape shaped by advanced AI systems. Thus, in this paper, we propose an end-to-end Machine Learning Framework to automate and strengthen cybersecurity operations. To demonstrate the feasibility of our approach, in this work-in-progress paper, we built and evaluated, a framework that consists of a machine learning (ML) model that automates the classifications of alerts generated by the Splunk Security Information and Event Management system (SIEM) to map adversary behaviors to MITRE ATT&CK stages. Our automated detection efforts are then fed into an LLM to generate an incident response report for cyber threat intelligence (CTI) and incident response teams. Our experimental results are intended to highlight the feasibility of combining adversary emulation with early real-time AI-assisted detection to strengthen organizational resilience (i.e. in Critical Infrastructure) against sophisticated campaigns.**

***Index Terms*—Artificial Intelligence (AI), Machine Learning (ML), Security Information and Event Management (SIEM), Advanced Persistent Threat (APT) Detection, MITRE ATT&CK**

I. Introduction

Defending organizational assets from nation-state hackers amplifies the challenges associated with defensive cybersecurity and necessitates the concurrent advancement of defensive frameworks. As such, to defend against AI-enabled APTs, system owners must also leverage AI to their advantage. This is especially important when protecting critical networks, such as critical infrastructure. Current tools and technologies can be expanded to support security operations. Thus, using frameworks such as MITRE ATT&CK and TTPs (tactics, techniques, and procedures) to track APT behavior, security teams can preemptively prepare and mitigate active and future threats to their critical networks. This approach also enables the conversion of cyber threat intelligence into strategic and actionable solutions for stakeholders.

Current applications of AI and large language models (LLMs) in intrusion detection do not adequately address the full-life cycle of APTs. Our research aims to fill this gap. By using AI to assist in the analysis of Splunk SIEM logs in real-time, we can also promote and increase the efficiency of real-time prevention and threat mitigation. We define the APT life cycle as follows: reconnaissance, initial access, credential theft, lateral movement, command and control (C2) compromise, and data exfiltration. Failure to detect sophisticated APTs early on can leave organizations vulnerable and exponentially increase risk while overwhelming security operation centers (SOC) and incident response teams.

Our current research scope aims to simulate AI-assisted defenses against all stages of the APT lifecycle including human-operated ransomware campaigns on critical infrastructure networks. Although the current evaluation focuses on a specific attack scenario, our framework and model are designed to be applicable to a variety of use cases. Extending ASOC-CA to broader scenarios and threats focused on critical infrastructure represents a possible direction for future work.

Finally, as we seek to propose solutions and map movements of adversaries, our proposed framework is intended to support and enhance defensive security operations. Our main contributions include: (1) a demonstration of AutoSOC Cyber Analyst, an AI-assisted defense against all stages of the APT lifecycle and (2) an evaluation of the performance of AutoSOC Cyber Analyst. ASOC-CA performs lightweight, real-time reporting while combining deterministic rules with machine learning to predict various MITRE ATT&CK stages.

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II. Related Works

A. SIEM and ML for Attack Detection

The current application of machine learning (ML) on APT detection primarily focuses supervised ML to triage SIEM alerts and host telemetry [1]. Machine learning techniques such as Random Forest and XGBoost have shown to reduce false positive alerts and achieve a near perfect accuracy of 98.6% and 99.6% respectively. Ali et al. tested their system in a batch process: first collecting alerts after an incident, refining them into a dataset, and then classifying them. Ravikumar proposed a similar scheme as Ali et al., creating a random forest model using the CSE-CIC-IDS2018 dataset to achieve a 98.7% accuracy [9].

There exists challenges to understand the reasoning behind AI decisions when it is mobilized in security operations [2]. The lack of transparency introduces hesitancy and delay for analysts to truly rely on autonomous systems. Structured and systematic real-time detection is important.

B. SIEM Rule to MITRE ATT&CK mapping

Another related line of work focuses on mapping security information to the MITRE ATT&CK framework. Much of this research focuses on Cyber Threat Intelligence reports or IDS/SIEM rules. For example, [5] introduces a Rule-ATT&CK Mapper (RAM) that converts SIEM rule descriptions into ATT&CK techniques and sub-techniques using an agentic LLM pipeline. RAM enriches SIEM rules statically - it does not identify active attacks or provide support during an intrusion. Meanwhile, SIEM applications for APT detection are challenged with alert fatigue and existing solutions are limited [3]. These solutions do not reflect the full APT lifecycle. However, due to rapidly changing attacks, vast opportunities to automate security operations still exist.

C. LLMs in Defensive Security Operations

Adding on, research on LLM support for SOC analysts found that AI-augmented systems consistently improve detection accuracy, reduce false positives, and accelerate response times. Srinivas et.al's systemic review portrayed the measurable operational benefits over traditional rule-based methods [4].

The most relevant example is CORTEX [6]. CORTEX introduces a multi-agent LLM architecture for high-stakes alert triage, where specialized agents analyze behavior sequences, gather evidence from enterprise tools, and synthesize an auditable triage decision. By basing decisions off real SIEM and identity data, CORTEX significantly reduces false positives compared to single-model baselines.

CORTEX focuses on multi-LLM investigations and conducts full-fidelity triage using multiple agents and external tool calls.

D. Existing SOC Automations

Lastly, Autonomous-SOC models do exist. Sri Lanka's Institute of Information Technology developed an AutoSOC that serves as a budget friendly SOC platform [7]. While this model does support security operations, it is a forensic tool-kit for post-incident activity. AutoSOC-CA is intended to support real-time detection and lifecycle phase mapping, especially as commonly advanced APTs can bypass traditional security measures and persist covertly in networks [3]. Thus, with new AI-systems, it is important to develop solutions that can foresee and identify the subtle cues of an APT.

III. System Overview

Evaluating ASOC-CA in a realistic operational setting, led to the emulation of the full detection pipeline inside a virtualized environment of a critical-infrastructure IT management network in GNS3 Network Emulator v2.2.54. This was deployed on a ASUS Zenbook Windows 11 64-bit PC with an Intel(R) Core(TM) i7-8565U CPU @ 1.80GHz processor and 16GB RAM. Figure 1 illustrates the topology used to capture live telemetry, execute attacks, and validate end-to-end alert classification. The environment models a segmented enterprise network with an internal management plane, a DMZ, and external adversarial traffic originating from the public Internet. However, the implementation of our project runs on a flattened network diagram under one local area network (LAN).

A. Network Architecture

The internal management segment contains a Windows Server domain controller (`dc-admin-ws16`), a file/application server, and an end-user workstation. These hosts would typically generate routine authentication, file-access, and system-level events. All internal nodes forward logs to a dedicated Splunk Enterprise server (`blueteamAI`), which acts as the central SIEM and telemetry aggregator for the entire topology.

A DMZ subnet contains four publicly reachable services: a VPN endpoint, a web gateway, a mail server, and an HTTP web server. These nodes were intentionally exposed to emulate typical externally facing critical-infrastructure assets. During testing, the `mailServer` played a central role as the target of brute-force authentication attempts, producing high-volume authentication and connection logs for ASOC-CA's RDP-brute-force evaluation dataset.

External adversarial activity originates from a simulated attacker host (`redteam-aptAI`), connected through the external firewall. This host was used to launch controlled brute-force attacks, allowing us to generate realistic malicious traffic while preserving full ground-truth labels for evaluation. Simulated network segmentation between the Internet, DMZ, and the internal switch ensures that the attack path and the visibility model mirror a real network environment that ASOC-CA can run on.

B. ASOC-CA Processing Pipeline

To evaluate the effectiveness of ASOC-CA, we launched our visualized environment with our attacker node and tested detection on an early phase of our attack lifecycle: Initial Access with a brute-force attempt over RDP. We ran this attack simulation using a python script 100-120 times to generate

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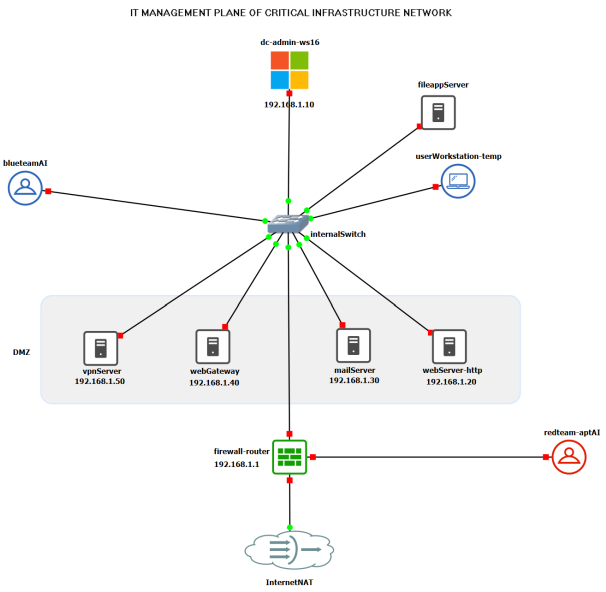


Fig. 1. Network Diagram

alerts and collect data. Our end-to-end pipeline consists of five stages:

1) **SIEM Log and Alert Ingestion:** All hosts were configured with Splunk Universal Forwarders to transmit Windows, Sysmon, and application logs to the Splunk Enterprise server. Splunk correlation searches generated alerts for anomalous authentication behavior (Figure 2), which ASOC-CA consumed in real time.

```
index=main sourcetype="WinEventLog:Security" (EventCode=4625 OR EventCode=4624)
| rex "Source Network Address:\s+(?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats count(eval(EventCode=4625)) AS total_failed, count(eval(EventCode=4624)) as
    total_succeed, values(src_ip) AS src_ips, values(_time) AS times BY host
| where (total_failed >= 3 AND total_succeed==1) OR (total_succeed >=3)
```

Fig. 2. Splunk Brute Force Query

2) **Feature Extraction:** Each alert received was ingested by ASOC-CA through an HTTP Event Collector (HEC) webhook, which was normalized and transformed into a structured feature vector containing source/destination IPs, host identifiers, outcome counts, temporal features, and protocol metadata.
3) **Machine Learning Classification:** A Random Forest (RF) classifier performs real-time, per-event inference. The model has been trained to predict the maliciousness of the activity and its corresponding MITRE ATT&CK Technique (e.g. T1110). Unlike prior SIEM-batch approaches, predictions are produced per-event as alerts arrive, enabling a low detection time.
4) **Deterministic MITRE ATT&CK Mapping:** The predicted ATT&CK labels are passed to a Deterministic Inference Engine. This engine follows a hierarchical logic: (1) If the RF Model predicts 'Malicious' with confidence > 0.9, the engine identifies the protocol and event code (e.g., RDP, ID 4625); (2) It maps these features to the primary Technique T1110 (Brute Force); (3) It cross-references a curated knowledge base to append vetted remediation steps (e.g., GPO paths for Account Lockout Policies) and common associated vulnerabilities. This ensures technical accuracy that purely probabilistic models cannot guarantee and reduces the risk of model hallucination.
5) **LLM-Generated Incident Reporting:** In the final stage, the fusion layer, ASOC-CA utilizes an LLM to create a human-readable incident summary describing the suspected attack stage, relevant indicators, affected hosts, and recommended response actions. The LLM ingests three distinct data sources: (1) raw telemtry logs from Splunk, (2) the ML classification labels, and (3) the determinsitic output. By incorporating these inputs, the LLM converts raw data into actionable intelligence, providing SOC analysts with a comprehensive summary of the "how", "what", and "how-to-fix" of a possible active intrusion.

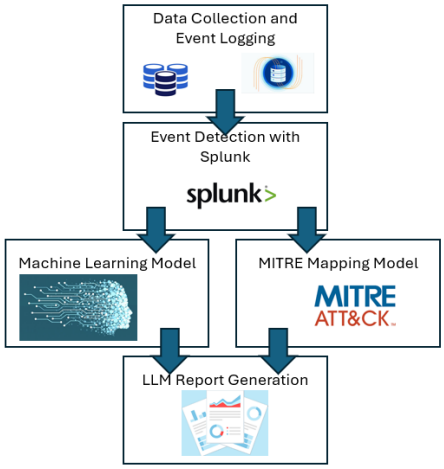


Fig. 3. High-level Architecture.

C. End-to-End Automation

This architecture enables ASOC-CA to operate as a fully automated blue-team assistant. Live logs from the GNS3 network trigger Splunk alerts. ASOC-CA classifies each alert, assigns a MITRE ATT&CK label, and produces a structured report within seconds. The tight integration of ML, rule-based reasoning, and LLM-based reporting allows ASOC-CA to function as an intelligent SOC copilot capable of both detection and interpretation during active intrusions.

IV. Dataset and Feature Engineering

A. Dataset

To support the development and evaluation of our intrusion-detection models, we generated a custom dataset focused on Remote Desktop Protocol brute-force behavior. We configured

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a Splunk alert to reliably capture authentication-related events associated with both legitimate and adversarial activity. Using this monitoring pipeline, we executed numerous controlled brute-force attacks originating from internet-based hosts to ensure realistic traffic patterns and variability in attacker behavior. All events were logged and exported through Splunk, producing a labeled dataset that reflects both successful and unsuccessful authentication attempts under diverse conditions. The dataset contains approximately 17% legitimate authentication, that consist of false positive alerts, and 83% malicious brute force attacks.

The training dataset is split into training and testing datasets with a 25:75 ratio. External validation was conducted using data that the model had not encountered during training, ensuring a realistic measure of its generalization performance. Because the dataset is imbalanced, relying solely on accuracy would provide an incomplete assessment; therefore, additional metrics as mentioned above serve to adequately validate the model's effectiveness across all classes.

B. Feature Selection and Engineering

The final dataset includes a set of features engineered to capture meaningful behavioral indicators relevant to RDP brute-force detection. Core network-level attributes such as *src_ip*, *host_ip*, and *host_device* provide context about the origin and target of each authentication attempt. Temporal characteristics enable modeling of attempt frequency and patterns typical of brute-force campaigns. Additionally, we include aggregated successes and failures features, which quantify the distribution of login outcomes over time and across attacking hosts. These features allow our model to distinguish normal user behavior from high-volume authentication failures indicative of credential-guessing attacks. Collectively, this engineered dataset provides a foundation for machine learning–based detection and analysis of RDP brute-force intrusions.

Our feature set was intentionally kept minimal to support real-time deployment within operational systems. While comprehensive datasets such as CSE-CIC-IDS2018 offer a richer representation of attack behavior and network dynamics, collecting upwards of 80 network features in real time introduces significant complexity, particularly in environments where cost, system overhead, or resource constraints are critical considerations. Splunk was selected in part because SIEM-based alerting requires minimal configuration and is widely adopted in settings where highly complex cybersecurity infrastructures are impractical to deploy or maintain effectively.

C. ML Models

To evaluate the feasibility of automated RDP brute-force detection, we trained and tested two complementary machine-learning models: an unsupervised K-Means clustering algorithm and a supervised Random Forest classifier. The K-Means model was selected to explore whether brute-force authentication patterns form naturally separable clusters without labeled data. In parallel, the Random Forest model was trained using labeled authentication outcomes, leveraging its robustness to nonlinear feature interactions and its ability to model complex behavioral signatures.

To ensure realism, both models were evaluated against traffic generated through true network simulations using the network topology shown in III, including controlled brute-force attacks using commonly leaked username/password pairs as well as randomized credential attempts. This approach allowed us to directly test model performance on practical adversarial behavior rather than synthetic or curated data alone.

D. Machine Learning Evaluation Metrics

Model performance was assessed using a broad suite of evaluation metrics reflecting both classification quality and predictive reliability. Accuracy and precision were measured along with counts of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) to quantify the system's ability to correctly identify brute-force events while minimizing misclassification.

In operational environments, false positives can trigger unnecessary alerts that disrupt routine activity, while false negatives represent undetected malicious behavior and therefore pose a direct security risk. Our objective is to minimize both types of errors to maintain system usability without compromising detection capability.

Accuracy measures the proportion of all correct predictions relative to the total number of evaluated instances. It provides a general indication of overall model performance across all classes. Formally, accuracy is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

where TP and TN denote true positive and true negative classifications, and FP and FN represent false positives and false negatives, respectively.

The True Positive Rate (TPR), also known as recall, quantifies the proportion of actual positive instances that the model correctly identifies. It reflects the model's ability to detect events of interest among all true positives. The metric is defined as:

$$\text{TPR} = \frac{TP}{TP + FN}$$

A corresponding measure for the negative class is the True Negative Rate (TNR), which is computed analogously. False Positive and Negative Rate (FPR/FNR) are similarly calculated.

Precision measures the proportion of correctly classified positive instances among all examples predicted as positive. The equation is given:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Collectively, these metrics provide a comprehensive assessment of both supervised and unsupervised approaches, demonstrating that machine learning can effectively detect RDP

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brute-force activity when evaluated against realistic network simulations.

E. Security Report Evaluation Metrics

To assess the quality and reliability of the LLM-generated security reports, we compared the executive summary that the model outputs against the structured input data provided to the model, including confidence level, source IP, host IP, host device, and aggregated outcomes to ensure model accuracy. The reports were further compared against the deterministic references input from the MITRE ATT&CK framework deterministic model. Recall was measured to determine the rate at which the inputted data was effectively output in the LLM report.

These inputs were also treated as the reference summaries for the purpose of qualitative evaluation using established natural language generation metrics, including ROUGE-1, ROUGE-2, and ROUGE-L, which measure lexical overlap and structural similarity between generated reports and reference summaries.

ROUGE-N measures n-gram overlap between a generated summary and a reference summary:

$$\text{ROUGE-N} = \frac{\sum_{g \in \text{Ref}} \min\left(\text{Count}_{\text{gen}}(g), \text{Count}_{\text{ref}}(g)\right)}{\sum_{g \in \text{Ref}} \text{Count}_{\text{ref}}(g)} \quad (1)$$

Where g is an n-gram, $\text{Count}_{\text{gen}}(g)$ is the number of times g appears in the generated report, and $\text{Count}_{\text{ref}}(g)$ is the number of times g appears in the reference summary.

ROUGE-L is based on the length of the longest common subsequence (LCS):

$$\text{ROUGE-L} = \frac{\text{LCS}(\text{gen}, \text{ref})}{\text{Length}(\text{ref})} \quad (2)$$

In addition to ROUGE-based metrics, we employed the BLANC metric to evaluate the functional quality of the LLM-generated reports as a whole. BLANC estimates summary effectiveness by measuring the performance improvement of a pre-trained language model when a generated summary is available during a language understanding task on the original document. Prior work has demonstrated that BLANC scores correlate strongly with human judgment and achieve comparable reliability to ROUGE-based evaluations. Importantly, BLANC does not require human-authored reference summaries, enabling fully automated and reproducible evaluation.

Formally, BLANC can be expressed as the difference between the model's performance with and without access to the summary:

$$\text{BLANC} = \mathcal{P}(D \mid S) - \mathcal{P}(D) \quad (3)$$

where D denotes the original document, S represents the generated summary, $P(D \mid S)$ is the language model's performance on a masked-token prediction task when conditioned on the summary, and P(D) is the baseline performance without the summary. We utilize BLANC to measure the accuracy of the overall report based on the summary.

We treat the generated executive summary as the generated summary and represent the overall LLM-generated report with and without the summary as a measure of the quality of the overall report.

V. Results and Discussion

A. ML Model Comparison

TABLE I
A comparison between Random Forest and K-Means Models

Model	ACC (%)	TPR (%)	TNR (%)	FPR (%)	FNR (%)	Precision (%)
Random Forest	98.1	98.9	92.9	1.06	7.14	98.9
K-Means	95.3	96.8	86.7	3.26	12.5	97.8

Across all evaluation metrics, the Random Forest model demonstrated strong performance, achieving high end-to-end accuracy (98.1%), precision (98.9%), and a notably high true positive rate (98.9%). Its low false positive rate (1.06%) and false negative rate (7.14%) indicate that the model rarely triggers unnecessary alerts while maintaining a reliable ability to detect malicious activity. In contrast, the K-Means clustering model achieved moderate accuracy (95.3%) and a true positive rate of 96.8%, with higher false positive (3.26%) and false negative (12.5%) rates than the Random Forest.

To establish a baseline for our model, we compare it to the performance of a standard rule-based SIEM (Splunk Correlation Searches) without ML intervention. In our experimental dataset, 17% of the alerts are legitimate authentication failures (False Positives) generated by the baseline Splunk correlation search. Without any further triage, a Tier-1 analyst would be required to investigate each alert. By implementing the ASOC-CA Random Forest classifier as a secondary triage layer, we drastically reduced this False Positive Rate from the 17% baseline to 1.06%, successfully filtering out over 93% of the baseline noise while maintaining a 98.9% True Positive Rate.

These results are comparable to prior work employing similar machine-learning architectures for intrusion detection. For example, Ali et al. report strong performance across multiple classifiers, with their Random Forest model achieving an accuracy of 98.5%, a true positive rate of 98.6%, and balanced false negative and false positive rates of 1.4% [1]. Their work did not include unsupervised learning. Ravikumar obtained 98.7% accuracy in their random forest model using the CSE-CIC-IDS2018 dataset [9]. It is important to note that both papers used a much larger set of features that likely account for the slight improvement of accuracy, but would make the models more challenging to apply in real time.

The models were evaluated in a virtualized critical infrastructure network based on the topology illustrated in Section III. These simulations included controlled brute-force attempts leveraging commonly leaked username-password combinations as well as fully randomized credential guesses. This methodology enabled us to assess model performance against genuine adversarial behavior rather than relying solely on synthetic or curated datasets. Throughout these experiments, the system consistently demonstrated strong detection capabilities while keeping false positive alerts to a minimum for both types

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of brute-force attempts. As part of this process, the LLM-based reporting component was also rigorously examined to verify its accuracy and overall effectiveness.

TABLE II
EVALUATION METRICS FOR THE LLM-GENERATED REPORTS

Metric	Recall (%)	ROUGE-1	ROUGE-2	ROUGE-L	BLANC
	85.3	0.347	0.159	0.216	0.125

The LLM-generated reports demonstrate strong performance across multiple evaluation metrics, indicating both factual completeness and linguistic coherence. The recall score of 85.3% suggests that the model successfully captures the majority of required security-relevant facts presented to the model, even in the summary. ROUGE-based metrics further show moderate lexical and structural alignment with reference content, with ROUGE-1, ROUGE-2, and ROUGE-L scores of 0.347, 0.159, and 0.216, respectively. These values indicate effective preservation of core concepts while allowing for paraphrasing and abstraction typical of generative reporting. Additionally, the BLANC score of 0.125 supports the conclusion that the data is effectively captured both within the larger report and in the executive summary. Collectively, these results demonstrate that the LLM produces informative reports that balance completeness with interpretability whilst minimizing potential hallucinations.

B. Discussion

A qualitative comparison of ASOC-CA against existing baselines and state-of-the-art frameworks is provided in Table III. While academic batch-processing models [1] and agentic triage systems [6] offer high accuracy, ASOC-CA is uniquely optimized for real-time Tier-1 operations by combining low-latency ML with deterministic remediation knowledge.

TABLE III
QUALITATIVE COMPARISON OF ASOC-CA VS. BASELINES

Feature	Rule-Based	Batch ML	Agentic AI	ASOC-CA
Detection Time	Real-time	Daily/Hourly	Minutes	**Seconds**
Logic Type	Static	Statistical	Probabilistic	**Hybrid**
Remediation	None	N/A	Halluc. Risk	**Vetted**
ATT&CK Mapping	Manual	N/A	Automated	**Hybrid**

Both of our models exhibit low false alarms and missed detections relative to typical intrusion-detection baselines, demonstrating their utility in detecting RDP brute-force behavior. Additionally, these techniques can be applied to multiple stages of the attack chain, and their combined use offers the potential for a more comprehensive, end-to-end detection pipeline. Overall, the Random Forest provides the strongest accuracy and reliability, while the K-Means model offers flexibility and responsiveness in dynamic operational settings.

Although less accurate overall, the K-Means model remains effective within the specific scenarios represented in our dataset. Importantly, its unsupervised nature allows it to adapt more quickly to changes in attacker behavior without requiring retraining, making it suitable for environments where new or evolving attack attempts are common and where reinforcement-style adaptation is beneficial.

The LLM-based system integrates human language, structured data, and evaluation metrics, including accuracy, precision, and error rates, with a report completion time of 45 seconds, a significant improvement over the minutes-to-hours typically required for manual reporting and analysis. Leveraging deterministic references from the MITRE ATT&CK framework enables the model to generate accurate and consistent determinations of vulnerabilities, attack techniques, and potential remediation steps. A calculated confidence interval is incorporated to enhance the reliability and quality of the final report.

At a larger scale, the model functions as an end-to-end assessment and reporting tool that can be easily adapted to diverse operational environments, network topologies, and exploit patterns, offering a flexible and scalable solution for automated cybersecurity analysis.

VI. LIMITATIONS AND FUTURE WORK

In this section, we discuss the limitations of the proposed framework and some corresponding future work. The dataset used is synthetic, relatively small and focuses exclusively on a single technique and stage of attack. However, we frame this as a controlled validation case study. Utilizing a GNS3-emulated environment provided a 'Gold Standard' ground truth, allowing for the rigorous calculation of the ROUGE and BLANC scores reported in Section VI. This approach establishes the architectural integrity of ASOC-CA before its expansion to the noisier, more diverse datasets planned for future work. Additionally, this model also relies on the Splunk toolset, which does not have open source capabilities.

For future work, we plan to expand the dataset and the model to include multiple stages and TTPs, build a baseline for evaluation, and finalize the LLM incident report feature to successfully enable the simulation of a full APT kill chain. We plan to also evaluate ASOC-CA under additional ML algorithms. By expanding automated threat mitigation and real-time response mechanisms and metrics, we can further enhance the practical applicability and robustness of the detection framework. Another possible direction of future work is to incorporate an Intrusion Prevention System (IPS) to automatically block and mitigate threats in real-time.

VII. CONCLUSION

In this work, we presented an end-to-end automated defense pipeline for detecting and reporting RDP brute-force attacks. By integrating a custom dataset, engineered features, and both supervised and unsupervised machine learning models, the framework demonstrated high accuracy and low false positive/negative rates in realistic network simulations. The inclusion of an LLM-based reporting component further enables rapid, interpretable analysis of vulnerabilities and attack patterns. Although the current implementation focuses on a specific attack type and stage, the framework provides a practical foundation that can be extended comprehensively to cover

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additional attack vectors and the full APT kill chain, offering a scalable and adaptable solution for real-world cybersecurity operations.

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